Ex. $A \to 0$ $\quad n=3,\ X_1X_2X_3$ $\quad$ Length

$B \to 10$

$C \to 11$

AAA	000	$l(A)+l(A)+l(A)$
AAB	0010	$l(A)+l(A)+l(B)$
⋮		⋮
BCA	10110	$l(B)+l(C)+l(A)$
⋮	⋮	⋮
CCC	111111	$l(C)+l(C)+l(C)$

$$\Rightarrow T(x)^n = \sum_{x_1}\sum_{x_2}\cdots\sum_{x_n} 2^{-(l(x_1)+\cdots+l(x_n))}$$

Length of total binary stream when $x_1x_2\cdots x_n$ is input

$\Rightarrow A_3 = 1$ (AAA)

$A_4 = 6$ (AAB, AAC, ABA, ACA, BAA, CAA)

$$T(x)^n = \sum 2^{-l(x_1x_2x_3)} = 1\cdot 2^{-3} + 6\cdot 2^{-4} + \cdots$$